Light

GCSE Computer Science — FDE Cycle and Systems (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A CPU processes instructions by repeating a continuous cycle.
Which three stages make up this cycle?

- ○ **A.** Fetch, decode, execute
- ○ **B.** Input, process, output
- ○ **C.** Read, calculate, write
- ○ **D.** Start, process, finish

Q2 (1 mark)

The Program Counter (PC) is an important CPU register.
What information does the PC hold?

- ○ **A.** The memory address of the next instruction to be fetched
- ○ **B.** The data value currently being processed
- ○ **C.** The number of programs running on the computer
- ○ **D.** The result of the previous calculation

Q3 (1 mark)

During the decode stage, the CPU processes the fetched instruction. What is the purpose of the decode stage?

- ○ **A.** To convert the instruction into a different programming language
- ○ **B.** To send the instruction back to RAM for storage
- ○ **C.** To display the instruction on the screen for the user
- ○ **D.** To interpret the instruction and determine what operation is required

Q4 (1 mark)

The execute stage completes the processing of an instruction. What happens during the execute stage?

- ○ **A.** The instruction is stored in permanent memory
- ○ **B.** The instruction is sent to the printer
- ○ **C.** The instruction is deleted from the computer
- ○ **D.** The instruction is carried out by the relevant CPU component

Q5 (1 mark)

The Memory Address Register (MAR) is used when accessing memory. What does the MAR store?

- ○ **A.** The address of the memory location being accessed
- ○ **B.** The data being written to memory
- ○ **C.** The instruction currently being decoded
- ○ **D.** The speed of the memory

Q6 (1 mark)

The Memory Data Register (MDR) is involved in data transfer. What does the MDR hold?

- ○ **A.** The address of the data in memory
- ○ **B.** The speed at which data is transferred
- ○ **C.** The data being transferred to or from memory
- ○ **D.** The size of the memory in bytes

Q7 (1 mark)

The CPU uses the address bus to communicate with memory. What travels along the address bus?

- ○ **A.** Control signals like read and write
- ○ **B.** Data values and program instructions
- ○ **C.** Memory addresses specifying which location to access

- ○ **D.** Electrical power for the memory chips

Q8 (1 mark)

The data bus is one of three main buses in a computer system.
What is the purpose of the data bus?

- ○ **A.** To carry data and instructions between the CPU and memory
- ○ **B.** To carry memory addresses from CPU to memory
- ○ **C.** To carry timing signals that synchronise operations
- ○ **D.** To carry error messages when programs crash

Q9 (1 mark)

Modern devices often contain embedded systems.
Which statement best describes an embedded system?

- ○ **A.** A computer system built into a device to control a specific function
- ○ **B.** A portable laptop computer that can be carried anywhere
- ○ **C.** A desktop computer hidden inside a cabinet
- ○ **D.** A gaming console connected to a television

Q10 (1 mark)

Embedded systems are found in everyday objects.
Which device contains an embedded system?

- ○ **A.** A traditional printed book
- ○ **B.** A wooden dining table
- ○ **C.** A digital washing machine with programmable cycles
- ○ **D.** A manual bicycle with no electronics

Q11 (1 mark)

The Von Neumann architecture is used in most modern computers.
What is the 'stored program concept' in Von Neumann architecture?

- ○ **A.** Programs must be printed on paper before they can run
- ○ **B.** Programs are stored on external USB drives only
- ○ **C.** Programs are permanently wired into the CPU hardware
- ○ **D.** Programs and data are both stored in the same main memory

Q12 (1 mark)

The Accumulator (ACC) is a register in the CPU.
What is the main purpose of the Accumulator?

- ○ **A.** To store the address of the next instruction
- ○ **B.** To count how many programs are running
- ○ **C.** To store the current instruction being executed
- ○ **D.** To store intermediate results and outputs from the ALU

Q13 (2 marks)

During the fetch stage, several operations occur.
What is the correct sequence of operations in the fetch stage?

- ○ **A.** CIR decoded → ALU activated → Result stored in ACC
- ○ **B.** Data loaded from RAM → Calculation performed → Result displayed
- ○ **C.** Address in PC copied to MAR → Memory accessed → Instruction placed in MDR → Copied to CIR → PC incremented
- ○ **D.** User input received → Input processed → Output displayed

Q14 (2 marks)

The Current Instruction Register (CIR) is used during instruction processing.
What is the role of the CIR in the FDE cycle?

- ○ **A.** To hold the memory address of data being processed
- ○ **B.** To hold the address of the next instruction to fetch
- ○ **C.** To hold the result of the most recent calculation
- ○ **D.** To hold the instruction currently being decoded and executed

Q15 (2 marks)

The control bus carries signals that coordinate system operations.
Which types of signals travel on the control bus?

- ○ **A.** Memory addresses and data values

- ○ **B.** Video signals and audio signals
- ○ **C.** Read/write commands, clock signals, and interrupt requests
- ○ **D.** File names and folder locations

Q16 (2 marks)

The Control Unit (CU) is active during the decode stage.
What does the Control Unit do during decode?

- ○ **A.** It performs arithmetic calculations on the data
- ○ **B.** It interprets the instruction in the CIR and generates appropriate control signals
- ○ **C.** It fetches the next instruction from memory
- ○ **D.** It stores the result of the instruction in memory

Q17 (2 marks)

The address bus connects the CPU to memory.
In which direction(s) do signals travel on the address bus?

- ○ **A.** Bidirectional: signals travel both from CPU to memory and from memory to CPU
- ○ **B.** The direction changes depending on the clock speed
- ○ **C.** Unidirectional: signals only travel from memory to CPU
- ○ **D.** Unidirectional: signals only travel from CPU to memory

Q18 (2 marks)

Embedded systems have characteristics that differ from general-purpose computers.
Which characteristics are typical of embedded systems?

- ○ **A.** High cost, large size, and ability to run any software
- ○ **B.** Frequent operating system updates and antivirus requirements
- ○ **C.** Multiple monitors, mechanical keyboards, and gaming graphics cards
- ○ **D.** Single dedicated function, low power consumption, and real-time operation

Q19 (2 marks)

The Program Counter changes during the FDE cycle.
When and why is the PC incremented?

- ○ **A.** During decode, to calculate the instruction length
- ○ **B.** After every clock pulse, regardless of the cycle stage
- ○ **C.** During execute, to count completed instructions
- ○ **D.** During fetch, to point to the next instruction in sequence

Q20 (2 marks)

The Von Neumann architecture has a performance limitation.
What causes this limitation?

- ○ **A.** The CPU can only perform one type of calculation

- ○ **B.** Instructions and data share the same memory and buses, preventing simultaneous access
- ○ **C.** The computer can only run one program at a time
- ○ **D.** Memory can only store a maximum of 1 kilobyte of data

Q21 (2 marks)

The data bus transfers information between CPU and memory.
In which direction(s) can data travel on the data bus?

- ○ **A.** Only from CPU to memory (writing only)
- ○ **B.** Only from memory to CPU (reading only)
- ○ **C.** Bidirectional: from CPU to memory (writing) and from memory to CPU (reading)
- ○ **D.** Data travels in circles around the bus continuously

Q22 (2 marks)

A modern car contains many embedded systems.
Which components in a car typically contain embedded systems?

- ○ **A.** The windscreen wipers, tyres, and brake pads
- ○ **B.** The fuel injection system, cruise control, and parking sensors
- ○ **C.** The seats, seatbelts, and floor mats
- ○ **D.** The paint finish, window glass, and door handles

Q23 (2 marks)

The CPU contains an internal clock.
What is the purpose of the CPU clock?

- ○ **A.** To display the current time on the screen
- ○ **B.** To synchronise and coordinate all CPU operations
- ○ **C.** To limit how long programs can run
- ○ **D.** To measure how much electricity the computer uses

Q24 (3 marks)

An instruction 'LOAD 300' means: load the value stored at memory address 300 into the Accumulator.
Memory address 300 contains the value 75.
Which sequence correctly describes the execute stage for this instruction?

- ○ **A.** The value 300 is placed directly into ACC; ACC now contains 300
- ○ **B.** Address 300 is placed in MAR; memory returns value 75 to MDR; value 75 is copied to ACC
- ○ **C.** The PC is set to 300; the next instruction is fetched from address 300
- ○ **D.** The value 75 is subtracted from 300; result 225 is stored in ACC

Q25 (3 marks)

A computer system has a 24-bit address bus.
What is the maximum amount of memory this system can address?

- ○ **A.** 24 bytes, because there are 24 address lines
- ○ **B.** 16,777,216 bytes (16 MiB), because 2^24 = 16,777,216 addressable locations
- ○ **C.** 24,000 bytes, because each bit can address 1000 bytes
- ○ **D.** 24 megabytes, because 24 bits equals 24 MB

Q26 (3 marks)

Before a fetch operation begins, the CPU registers contain:
- PC = 500
- MAR = 200
- MDR = 99
- CIR = old instruction

After fetch completes, what will the PC contain?

- ○ **A.** 500, because PC does not change during fetch
- ○ **B.** 501, because PC is incremented to point to the next sequential instruction
- ○ **C.** 200, because PC takes the value from MAR
- ○ **D.** 99, because PC takes the value from MDR

Q27 (3 marks)

A system has a 16-bit data bus and needs to transfer 128 bytes from memory to the CPU.
How many bus transfer operations are required?

- ○ **A.** 128 transfers, because each byte needs a separate transfer
- ○ **B.** 8 transfers, because 128 ÷ 16 = 8

- ○ **C.** 64 transfers, because 16 bits = 2 bytes, and 128 ÷ 2 = 64
- ○ **D.** 16 transfers, because the bus is 16 bits wide

Q28 (3 marks)

A manufacturer is designing a digital fitness tracker worn on the wrist. Why is an embedded system more appropriate than a general-purpose computer for this device?

- ○ **A.** An embedded system can display high-resolution videos better than a computer
- ○ **B.** An embedded system can run office software like word processors and spreadsheets
- ○ **C.** An embedded system offers small size, low power consumption, and dedicated functionality suitable for a wearable device
- ○ **D.** An embedded system provides faster internet browsing than a laptop

Q29 (3 marks)

An instruction 'STORE 400' means: store the value in the Accumulator to memory address 400.
The ACC currently contains 60.
Which sequence correctly describes the execute stage for this instruction?

- ○ **A.** Address 400 is placed in MAR; value 60 from ACC is placed in MDR; MDR contents written to memory at address in MAR
- ○ **B.** Value 400 is copied into ACC; ACC now contains 400
- ○ **C.** Memory address 400 is read into MDR; MDR value copied to ACC

○ **D.** PC is set to 400; execution continues from address 400

Q30 (3 marks)

A program contains these four instructions:
• Address 200: LOAD 500
• Address 201: ADD 501
• Address 202: ADD 502
• Address 203: STORE 503
How many complete FDE cycles are needed to execute this entire program?

○ **A.** 1 cycle, because all instructions are processed together

○ **B.** 4 cycles, because each instruction requires one complete fetch-decode-execute cycle

○ **C.** 12 cycles, because each of the three FDE stages counts as a separate cycle

○ **D.** 8 cycles, because fetch and execute each need their own cycle

Total: **55** marks • Time: **30 min**

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